

Image Processing (2)

Shin'ichi Satoh

Review of the last lecture

- Image representation, model, and process.
- Convincing reasons in terms of human visual sensation and/or physical imaging.
- As an example of human visual sensation, color sensation was visited.

Today's topics

- Physical imaging system will be studied
- Radiometry (radiance/irradiance)
- Bidirectional Reflectance Distribution Function (BRDF)
- Surface reflectance properties (Lambertian)
- Reflectance map
- Photometric stereo

Image Formulation

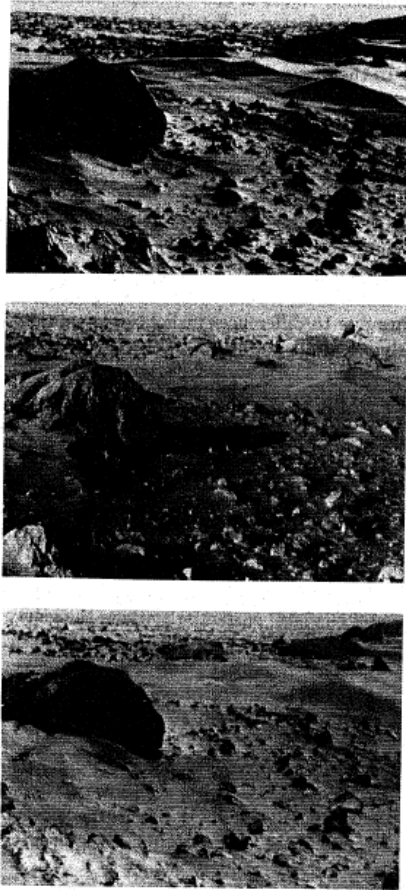


Figure 10-1. The appearance of a surface depends greatly on the lighting conditions. We have to understand how images are formed if we are to recover information about the surface from one or more images. Shown here are three views of the surface of Mars taken by Viking Lander I. (Pictures IPL PIC ID 77/03/06/134059, 77/02/09/172643, and 77/02/05/042935, courtesy of the National Space Science Data Center, Greenbelt, Maryland.)

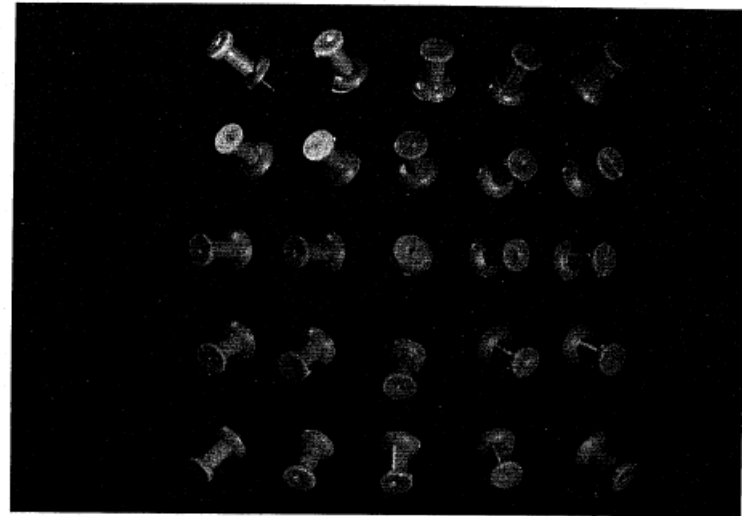


Figure 10-2. The appearance of an object depends greatly on its attitude in space relative to the viewer. Not only does the outline vary, but the brightness pattern within the silhouette changes.

Imaging Process (pinhole camera)

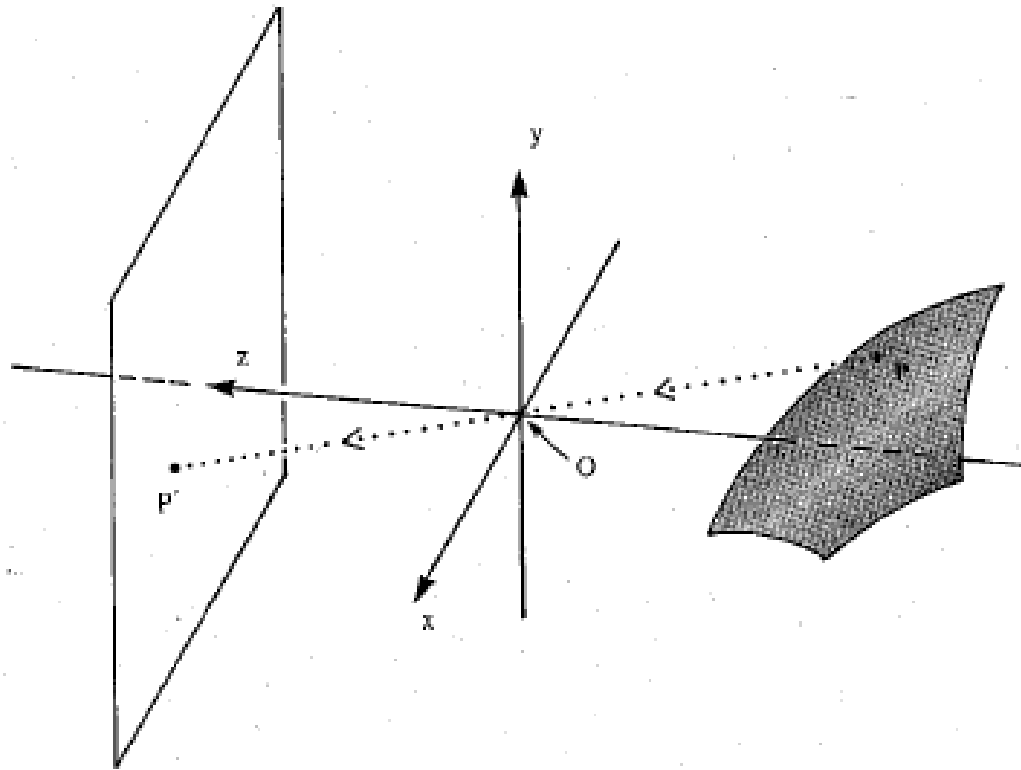


Figure 2-1. A pinhole camera produces an image that is a perspective projection of the world. It is convenient to use a coordinate system in which the xy -plane is parallel to the image plane, and the origin is at the pinhole O . The z -axis then lies along the optical axis.

Radiometry

Irradiance

Radiance

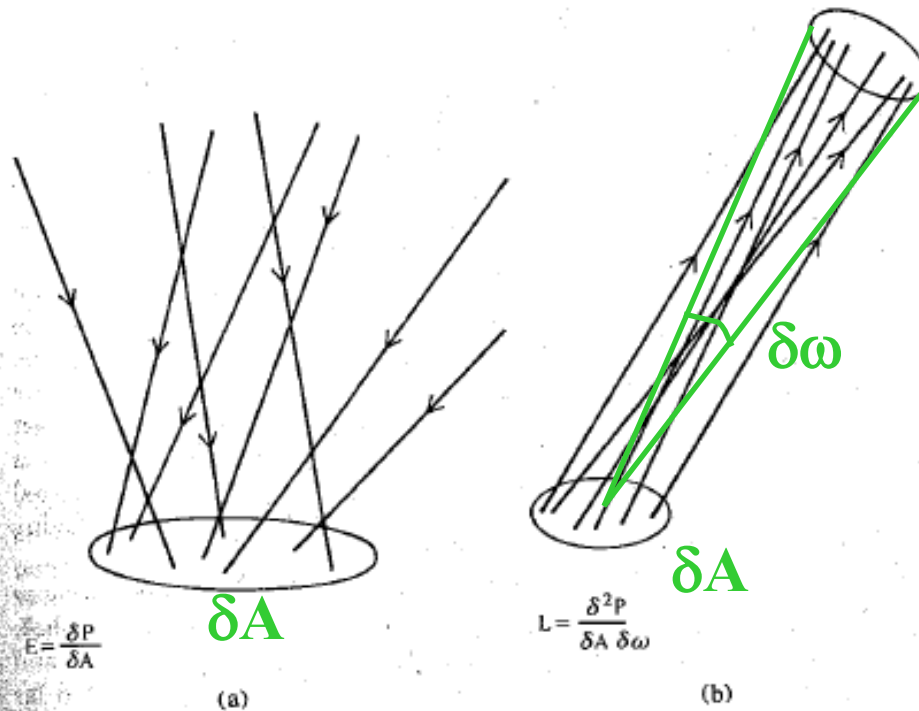


Figure 2-3. (a) Irradiance is the power per unit area falling on a surface. (b) Radiance is the power emitted per unit area into a cone of directions having unit solid angle. The term *brightness* is used informally for both concepts.

Imaging Process (Lens)

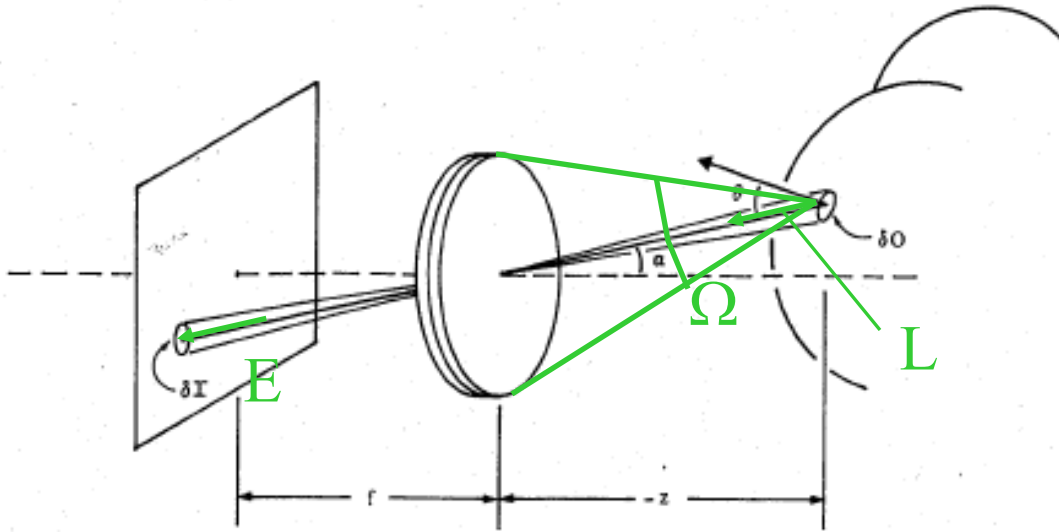


Figure 10-4. To see how image irradiance is related to the radiance of the surface, we must determine the size of the region in the image that corresponds to the patch on the surface.

$$\Omega = \frac{\pi}{4} \left(\frac{d}{z} \right)^2 \cos^3 \alpha$$

$$\delta P = L \delta O \Omega \cos \theta$$

$$\begin{aligned} E &= \frac{\delta P}{\delta I} = L \frac{\delta O}{\delta I} \frac{\pi}{4} \left(\frac{d}{z} \right)^2 \cos^3 \alpha \cos \theta \\ &= L \frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \end{aligned}$$

Bidirectional Reflectance Distribution Function (BRDF)

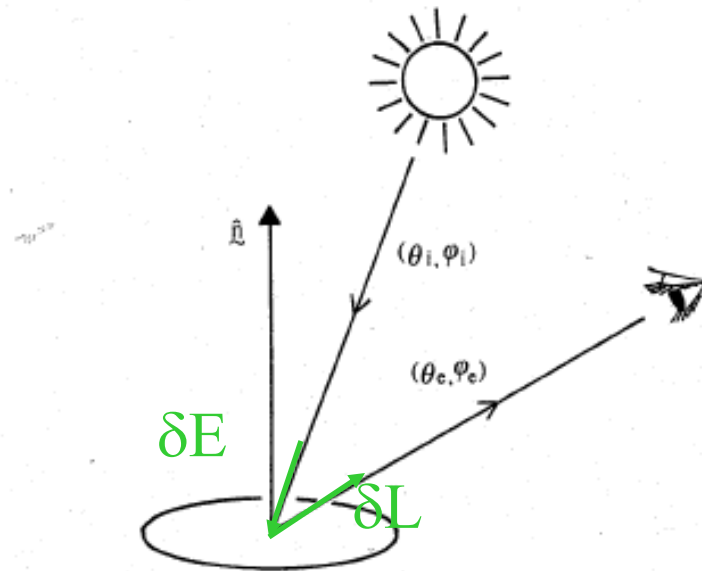


Figure 10-7. The bidirectional reflectance distribution function is the ratio of the radiance of the surface patch as viewed from the direction (θ_e, ϕ_e) to the irradiance resulting from illumination from the direction (θ_i, ϕ_i) .

BRDF:
$$f(\theta_i, \phi_i; \theta_e, \phi_e) = \frac{\delta L(\theta_e, \phi_e)}{\delta E(\theta_i, \phi_i)}$$

Surface Reflectance Properties

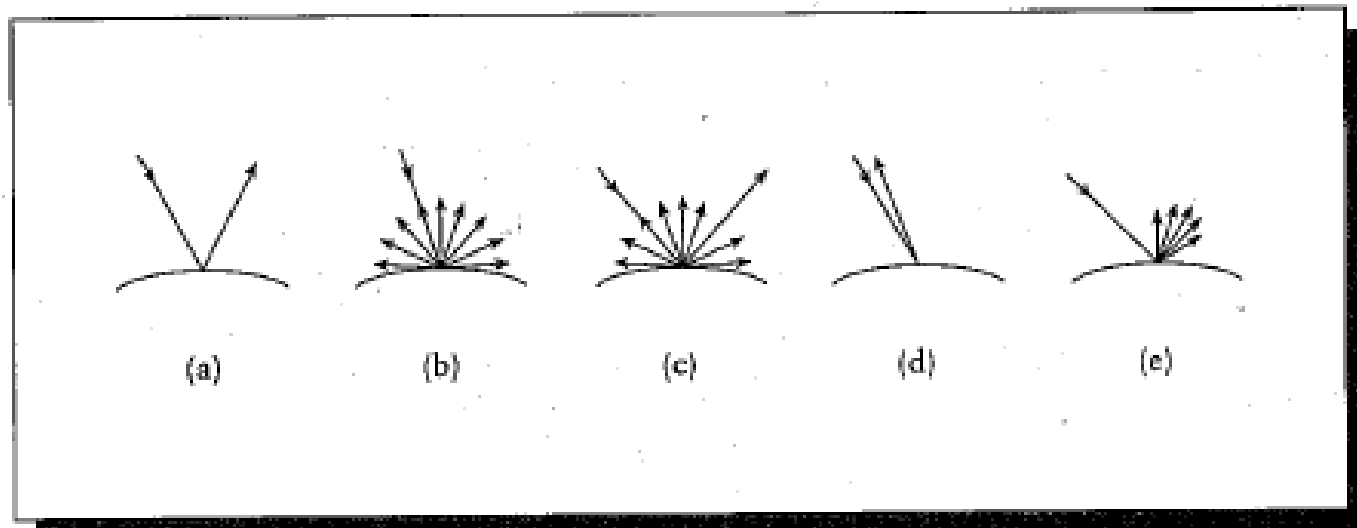


FIGURE 11.11

Different forms of reflection. (a) Specular. (b) Diffuse. (c) Mixed. (d) Retro-reflection. (e) Gloss.

Physical Model of Specular and Diffuse Reflection

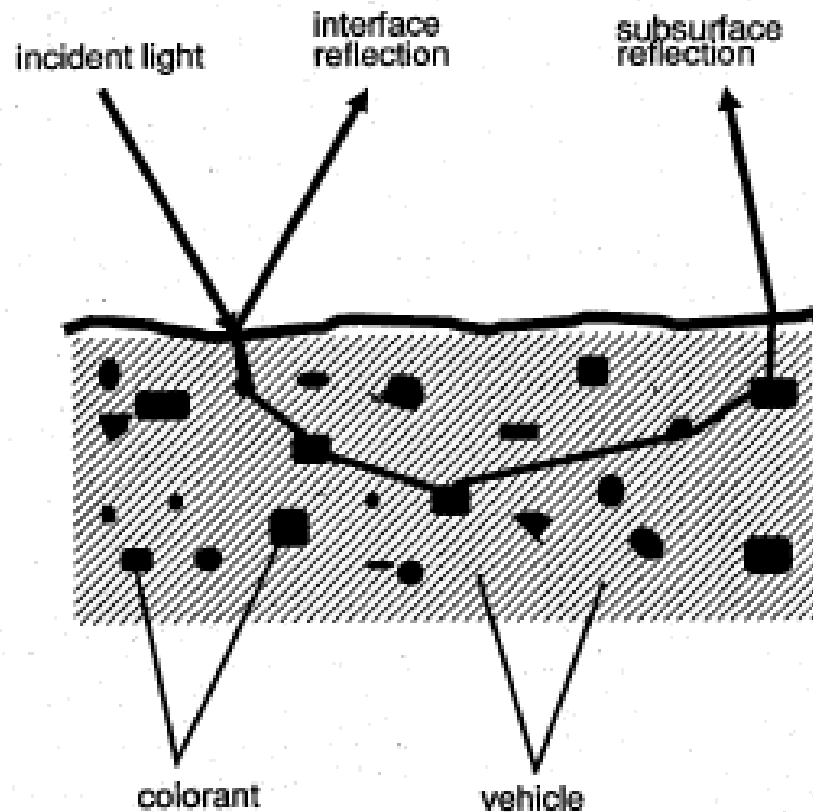


Fig. 1. Light reflected from a surface is modeled as the sum of two components. Each component represents a different light path.

Surface Orientation as Gradient

$$p = \frac{\partial z}{\partial x}$$

$$q = \frac{\partial z}{\partial y}$$

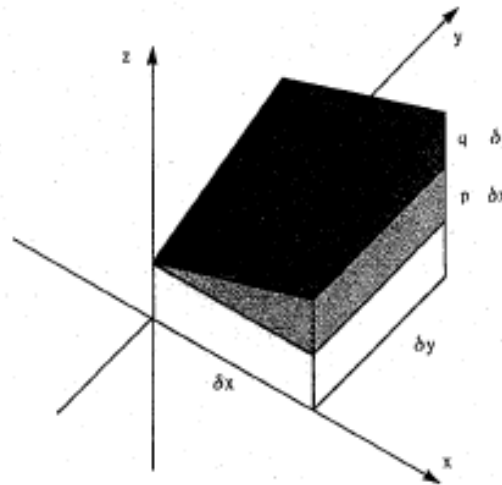


Figure 10-11. Surface orientation can be parameterized by the first partial derivatives p and q of the surface height z .

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{|\mathbf{n}|} = \frac{(-p, -q, 1)^T}{\sqrt{1 + p^2 + q^2}}$$

Reflectance Map (Lambertian)

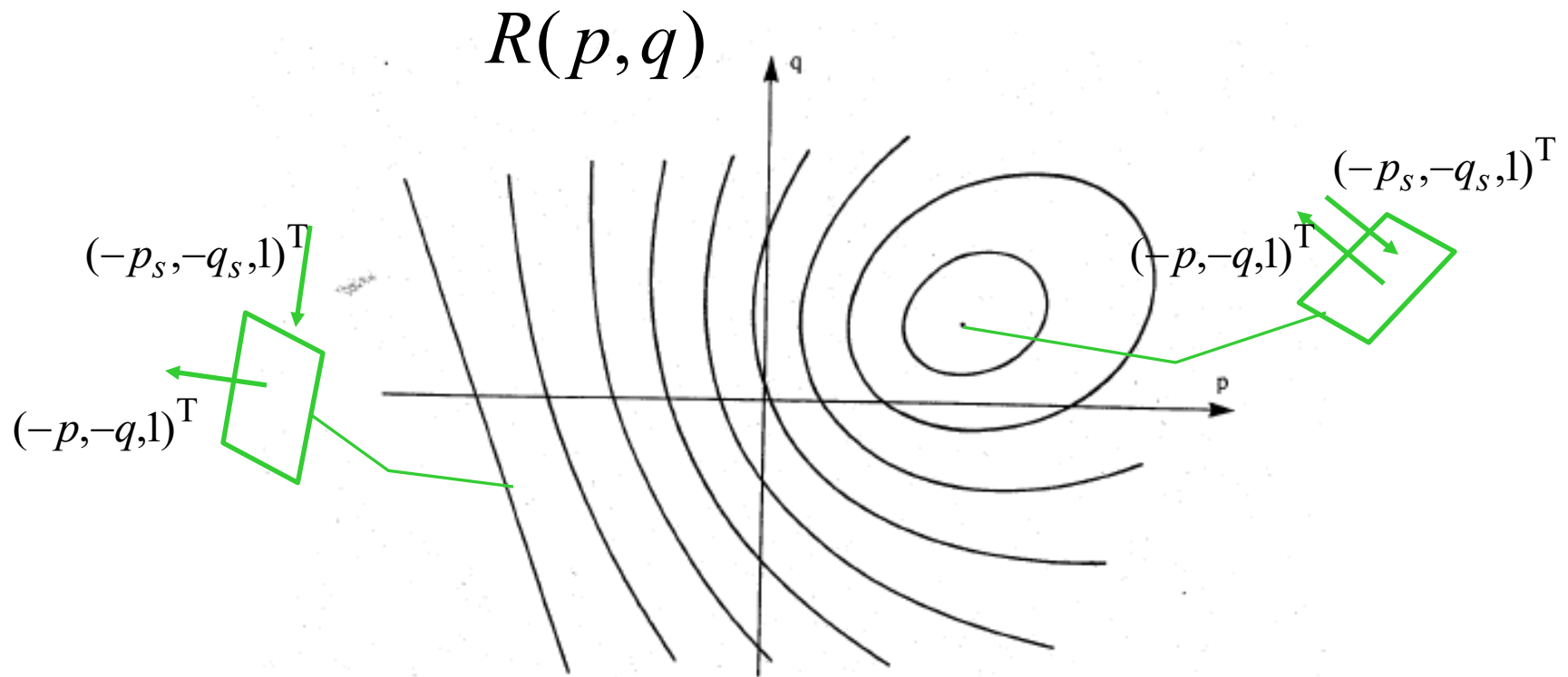


Figure 10-13. The reflectance map is a plot of brightness as a function of surface orientation. Here it is shown as a contour map in gradient space. In the case of a Lambertian surface under point-source illumination, the contours turn out to be nested conic sections. The maximum of $R(p, q)$ occurs at the point $(p, q) = (p_s, q_s)$, found inside the nested conic sections, while $R(p, q) = 0$ all along the line on the left side of the contour map.

Reflectance Map (Diffuse+Gloss)

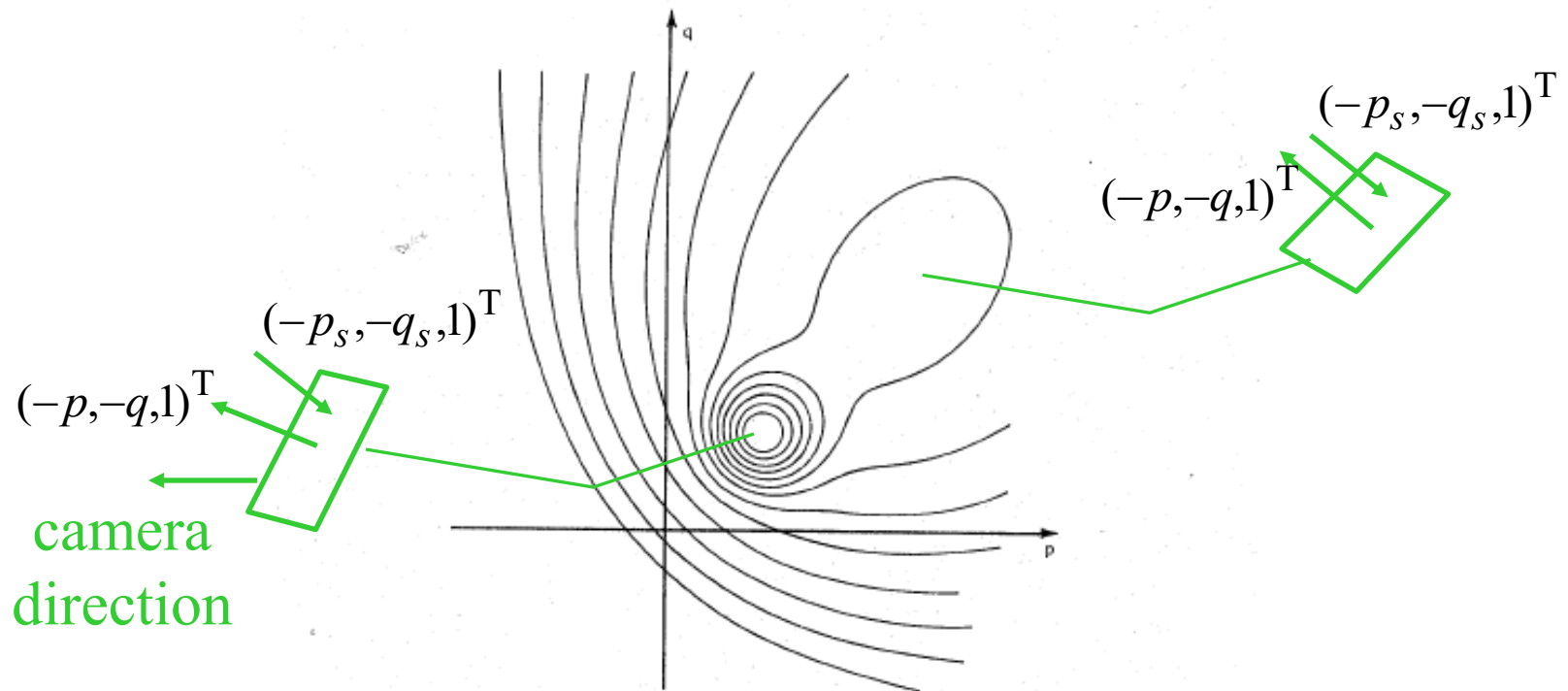


Figure 10-15. Many real surfaces combine diffuse and glossy components of reflection. The glossy component comes from reflection at the interface between air and the surface material, while the diffuse component can be traced to the component of light that has penetrated some distance into the surface, to be refracted and reflected until it reemerges from the surface layer. The reflectance map for such a material illuminated by a point source can have two peaks, corresponding to the surface orientations that maximize each of the two different types of reflection.

Usage of Reflectance Map: Shaded Graphics production

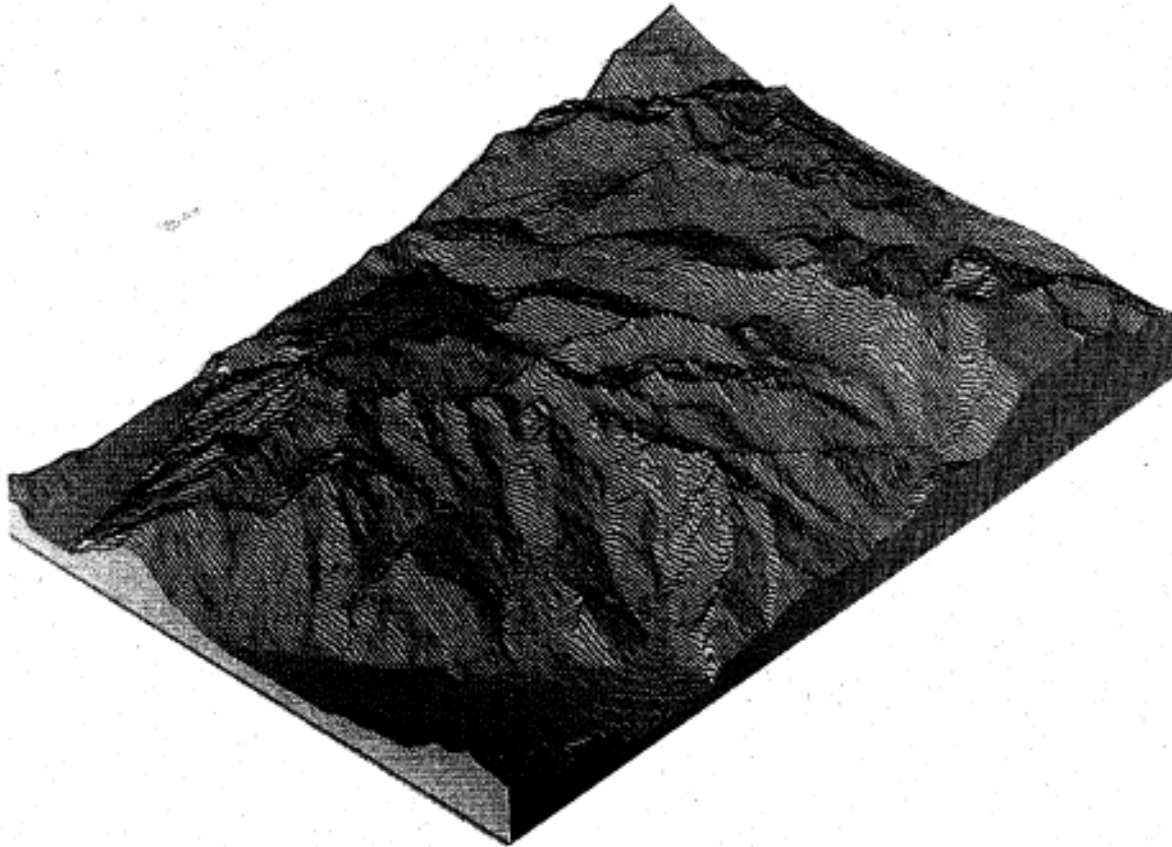


Figure 10-18. A block diagram made from a depth map of the surface of a mountainous region of the earth. (Digital terrain model kindly provided by Kurt Brassel.)

Usage of Reflectance Map: Shaded Graphics production

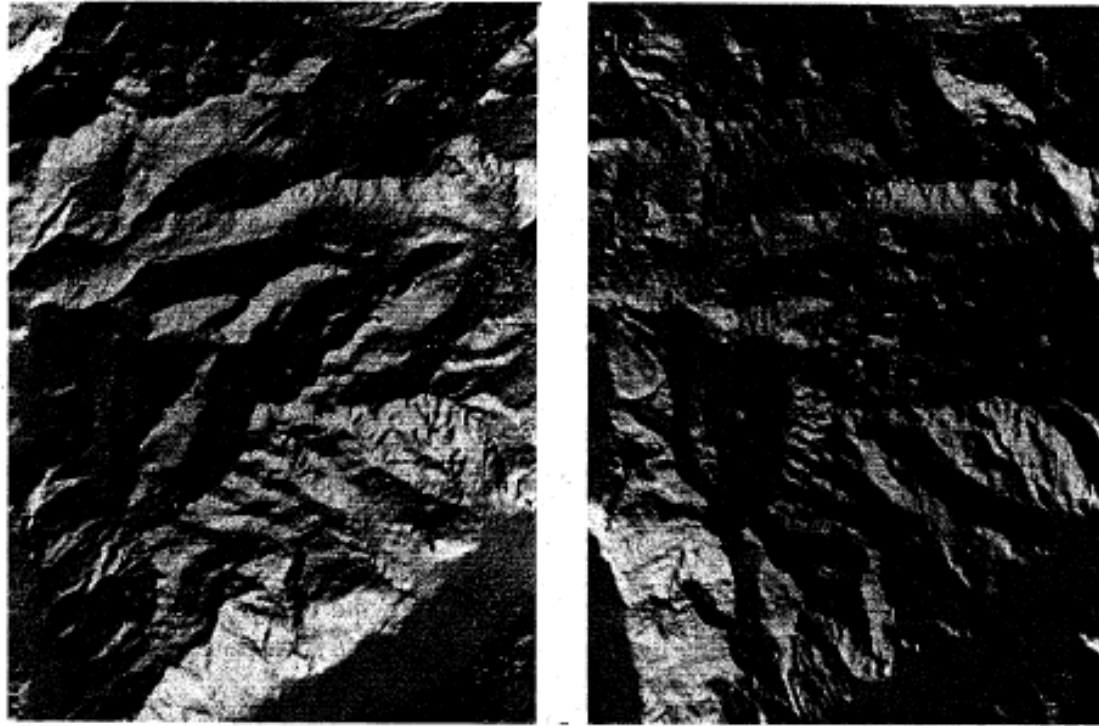


Figure 10-19. Shaded views of the surface shown in the previous figure. Here we are looking down on the surface from above. In the figure on the left, the light is assumed to come from the lower right, as is appropriate for a point on the earth north of the equator in the morning. In the figure on the right, the light is assumed to come from the lower left, corresponding to lighting conditions in the afternoon.

Photometric Stereo

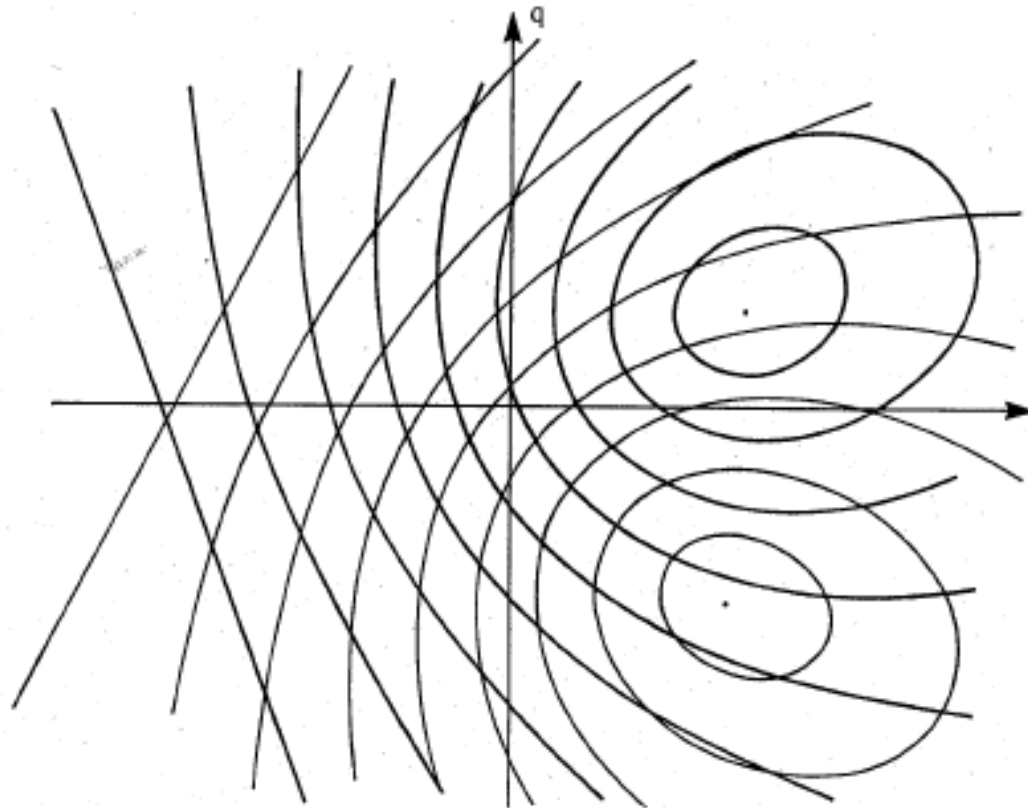


Figure 10-21. In the case of a Lambertian surface illuminated successively by two different point sources, there are at most two surface orientations that produce a particular pair of brightness values. These are found at the intersection of the corresponding contours in two superimposed reflectance maps.

Resultant Needle Diagram

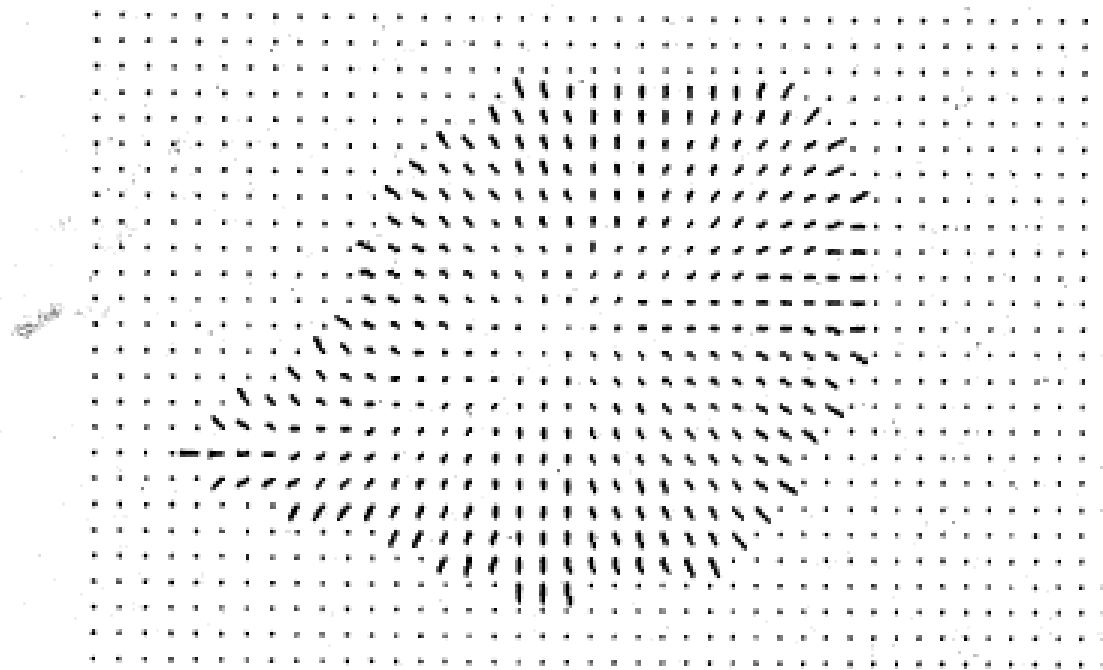


Figure 11-11. Needle diagram calculated by the iterative scheme under the assumption that the reflectance map is $\sec \theta_i$. (The surface orientation data are actually available on a finer grid; they are sampled coarsely here for display purposes.) The needle diagram is the estimate of the shape of the surface of the resin droplet shown in the previous figure. (Figure kindly provided by Katsushi Ikeuchi.)

Chair Assignment

5/9 Md. Hasanuzzaman

5/16 Chikahito Nakajima

5/23 Kuruma

(5/30 no class)

6/6 Kawasaki

6/13 Pen Kimveasna

6/20 Kiatisevi Pattara

6/27 reserved

7/4 reserved